

Operator Discovery Report: descriptor_aware_scaffold_selector

- Operator type: scaffold_selector.
- Host scaffold: nearest_neighbor_2opt.
- Core idea: Choose the most transfer-robust baseline based on instance structure descriptors (two-cluster bottlenecks, grid-likeness, elongated spreads, and nearest-neighbor trap tendency), with a safe fallback to nearest_neighbor_2
- Novelty classification: descriptor_based_scaffold_selector.
- Rediscovery signature: scaffold_selector:random_restart_2opt:clustered_local_search:clustered_local_search.
- Surviving candidate: False.
- Problem class where it helps: heldout_tsplib, two_cluster_bottleneck_tsp.

Pseudocode

- Read instance descriptors to classify the layout family.
- Dispatch the instance to the preferred scaffold for that structure class.
- Fall back to a safe default scaffold when the structure is ambiguous.

Complexity

- Non-empty lines: 13.
- Complexity score: 0.46.
- Fingerprint: 1d19d0cb1e42320b.

Why It Should Help

- The operator selects among simple solvers from instance descriptors, so any gain should come from a reusable structure-to-scaffold mapping.

Failure Cases

- Underperforms on grid_like_tsp with gap delta 0.170114.
- Underperforms on nearest_neighbor_trap_tsp with gap delta 0.034212.
- Underperforms on uniform_euclidean with gap delta 0.013605.
- Does not transplant cleanly into cheapest_insertion_2opt.
- Runtime inflation is high at 1.105281.

Transplant Results

- nearest_neighbor_2opt gap delta -0.059449, runtime delta 274.928857, distance-eval delta 112789.0.
- cheapest_insertion_2opt gap delta 0.047873, runtime delta -4660.927757, distance-eval delta -6648269.0.
- random_restart_2opt gap delta -0.039123, runtime delta 143.455186, distance-eval delta 98603.428571.
- sparse_three_opt gap delta -0.058163, runtime delta 331.502629, distance-eval delta 109367.714286.
- clustered_local_search gap delta -0.073402, runtime delta 159.034029, distance-eval delta 98934.428571.

Instance-Family Results

- heldout_tsplib operator gap 0.17836 vs baseline 0.235058.
- clustered_tsp operator gap 0.113212 vs baseline 0.110853.
- elongated_corridor_tsp operator gap 0.006647 vs baseline 0.003204.
- grid_like_tsp operator gap 0.27681 vs baseline 0.106696.
- nearest_neighbor_trap_tsp operator gap 0.034212 vs baseline 0.0.
- two_cluster_bottleneck_tsp operator gap 0.001685 vs baseline 0.109057.
- uniform_euclidean operator gap 0.013605 vs baseline 0.0.

Pareto Results

- Gap delta: -0.014035.
- Runtime inflation: 1.105281.
- Distance-evaluation inflation: 11.887779.
- Same-gap-faster flag: False.
- Better-gap-same-runtime flag: False.

Ablation Results

- disabled mean gap 0.121317 and delta vs full operator 0.013723.
- flattened_selector mean gap 0.121317 and delta vs full operator 0.013723.
- shuffled_selector mean gap 0.075559 and delta vs full operator -0.032035.